Recap : Normal Equations

Homework 2 out today (processing a RAW image). Lecture topics : least squares problems, orthogonality, polynomial regression. **Common theme today : orthogonality.**

Key formulas

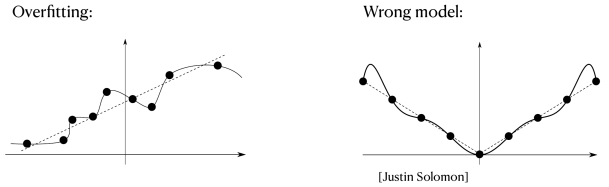
The "best" solution (in the L_2 sense) of a tall system is given by

$$\mathbf{A}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b}$$

Key idea

The normal equations yield a symmetric *quadratic* linear system, whose size is given by the number of columns of $\mathbf{A}$. Geometrically, $\mathbf{y} = \mathbf{A}\mathbf{x}$ is the point of $\operatorname{span}(\mathbf{A})$ closest to $\mathbf{b}$: the residual $\mathbf{r}(\mathbf{x})$ is perpendicular to it.

Regression and its limitations : overfitting, wrong model.



Warning

Okay, so what's the catch?

$$\operatorname{cond}(\mathbf{A}^T\mathbf{A}) \approx \operatorname{cond}(\mathbf{A})^2$$

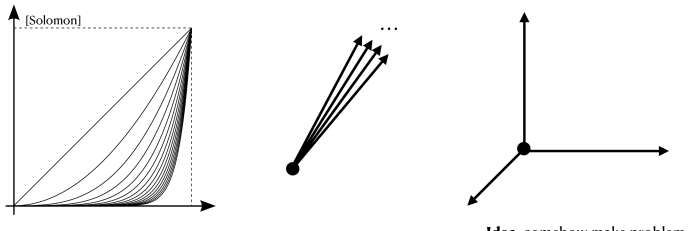
The condition number of a problem expressed using the normal equations is often *much worse* than the condition number of the original problem.

Key idea

One of our goals for today : can we solve the same problem, but without ever building $\mathbf{A}^T\mathbf{A}$?

Orthogonality

The issue with the Vandermonde matrix (geometric interpretation) : monomials of different powers look more and more alike, so the columns are almost parallel. **Idea** : somehow make the problem more "orthogonal".



Definition

INNER PRODUCT (measures a projection)

$$\langle \mathbf{a}, \mathbf{b} \rangle = \mathbf{a} \cdot \mathbf{b} = \mathbf{a}^T\mathbf{b} = \|\mathbf{a}\| \, \|\mathbf{b}\| \cos\theta$$

- If $\mathbf{a}$ has unit length, $\mathbf{a}^T\mathbf{a} = 1$.
- If $\mathbf{a}$ and $\mathbf{b}$ are perpendicular, $\mathbf{a}^T\mathbf{b} = 0$.

Gram-Schmidt Orthogonalization

Definition

GRAM-SCHMIDT STEP

$$\hat{\mathbf{y}} := \mathbf{y} - \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\langle \mathbf{x}, \mathbf{x} \rangle}\mathbf{x}$$

Then $\langle \mathbf{x}, \hat{\mathbf{y}} \rangle = 0$: remove from $\mathbf{y}$ its projection on $\mathbf{x}$.

Example

$\mathbf{x} = (1,2,3)^T, \mathbf{y} = (4,5,6)^T : \langle \mathbf{x}, \mathbf{y} \rangle = 1{\cdot}4 + 2{\cdot}5 + 3{\cdot}6 = 32.$

Definition

INNER PRODUCTS ARE MORE GENERAL

$$\langle \mathbf{x}, \mathbf{y} \rangle := \sum_{i=1}^{n} x_i y_i \qquad \langle f, g \rangle := \int_a^b f(x)g(x)\,\mathrm{d}x$$

Gram-Schmidt Applied to Polynomials

Key formulas

Everything defined on $[-1,1]$. Input (monomials) : $f(x) = 1$, $g(x) = x$, $h(x) = x^2$.

$$\langle f, g \rangle = \int_{-1}^{1} 1 \cdot x \,\mathrm{d}x = \left[\tfrac{1}{2}x^2\right]_{-1}^{1} = 0$$

$$\langle g, h \rangle = \int_{-1}^{1} x \cdot x^2 \,\mathrm{d}x = \left[\tfrac{1}{4}x^4\right]_{-1}^{1} = 0$$

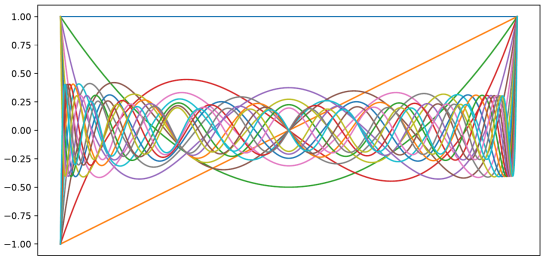
$$\langle f, h \rangle = \int_{-1}^{1} 1 \cdot x^2 \,\mathrm{d}x = \left[\tfrac{1}{3}x^3\right]_{-1}^{1} = \tfrac{2}{3}$$

$$\hat{h}(x) = h(x) - \frac{\langle f, h \rangle}{\langle f, f \rangle} f(x) = x^2 - \frac{1}{3}$$

Definition

LEGENDRE POLYNOMIALS

- Span the same domain as monomials, but they are **orthogonal to each other**!
- Problems with poor condition number disappear completely!
- Use them when you need to fit a smooth function to some data.



Runge Function and Chebyshev Points

Demo : the Runge function, fitted by polynomials through uniformly spaced points.

Definition

TAYLOR'S THEOREM
Approximation of a function around a point a. The residual of the approximation is equal to the $k{+}1^{\text{st}}$ derivative *somewhere* between the expansion point and where the function is evaluated. What about the error away from this point ?

Key formulas

Taylor's theorem for polynomials (approximation around a sequence of points x_i used for the fit) :

$$R_{k-1}(x) = \frac{f^k(y)}{k!}\prod_{i=1}^{k}|x - x_i|$$

Idea : try to make $\prod_{i=1}^{k}|x-x_i|$ small. **Optimal** set of points that achieve this (**Chebyshev points**) :

$$x_i = \cos\left(\frac{2k-1}{2n}\pi\right)$$

Key idea

Demo, Runge function part II : using Chebyshev points instead of uniformly spaced ones.

The QR Decomposition

Definition

QR FACTORIZATION

$$\mathbf{A} = \mathbf{QR}$$

- *Different flavors exist* : Gram-Schmidt, Givens rotations, **Householder transformations**.
- Follows the previous approach of using (pre-)computation to transform a "difficult" linear system into one that has a special structure, making it "easy".
- Compared to LU, one triangular factor was replaced by an **orthogonal matrix**.

Key formulas

$$\mathbf{Q} = \begin{pmatrix} | & & | \\ \mathbf{q}_1 & \cdots & \mathbf{q}_n \\ | & & | \end{pmatrix}$$

$$\mathbf{Q}^T\mathbf{Q} = \begin{pmatrix} \mathbf{q}_1^T\mathbf{q}_1 & \cdots & \mathbf{q}_1^T\mathbf{q}_n \\ \vdots & & \vdots \\ \mathbf{q}_n^T\mathbf{q}_1 & \cdots & \mathbf{q}_n^T\mathbf{q}_n \end{pmatrix} = \mathbf{I}$$

Key idea

Orthogonal transformations preserve distances, angles. They do **not** amplify error!

Definition

ISOMETRY
A transformation that preserves angles & distances. Rotate, mirror : isometric. Shear (and scaling) : **not** isometric.

Warning

The problem : least squares problems seek the solution in $\text{span}(\mathbf{A})$ that is *closest* to $\mathbf{b}$. If we non-isometrically transform the problem (as the LU decomposition does), then that actually changes the answer…

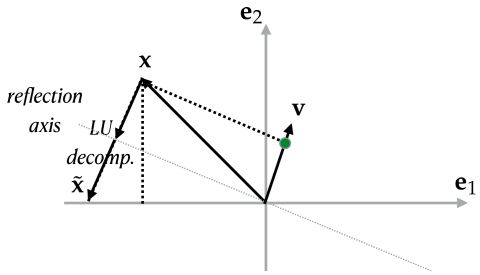
Householder Reflections

Key idea

The challenge : like LU, make $\mathbf{A}$ upper triangular ("R" as in *right triangular*, the same as upper-triangular), column by column :

$$\mathbf{Qx} = \mathbf{Q}\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} \tilde{x}_1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

Goal : zero out all elements of $\mathbf{x}$ except for the first. **Challenge** : need to do this only using rotations or reflections.



Definition

HOUSEHOLDER REFLECTION
Reflect $\mathbf{x}$ across the axis perpendicular to $\mathbf{v}$ (assuming $\|\mathbf{v}\| = 1$) :

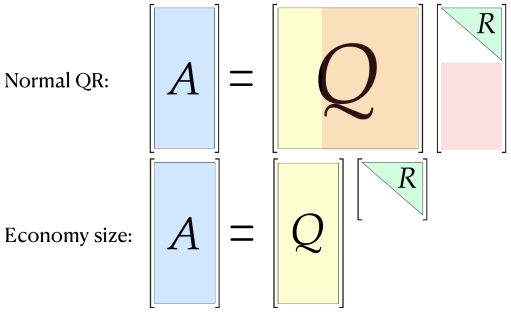
$$\tilde{\mathbf{x}} = \mathbf{x} - 2(\mathbf{v}^T\mathbf{x})\mathbf{v}$$

General form :

$$\tilde{\mathbf{x}} = \mathbf{Qx}, \quad \text{where } \mathbf{Q} = \mathbf{I} - 2\frac{\mathbf{vv}^T}{\mathbf{v}^T\mathbf{v}}$$

Using the QR Factorization

The end result for a tall $\mathbf{A}$: square orthogonal $\mathbf{Q}$, and $\mathbf{R}$ upper triangular on top of zeros. **"Economy size"** : drop the columns of $\mathbf{Q}$ that only multiply these zeros.



Key formulas

$$\mathbf{x} = \mathbf{R}^{-1}(\mathbf{Q}^T\mathbf{b})$$

Summary

- Be mindful to **not** compute an inverse of $\mathbf{R}$. Instead, solve the upper-triangular system $\mathbf{Rx} = \mathbf{Q}^T\mathbf{b}$ via back-substitution!
- More expensive to compute than LU factorization ($\sim$2X). Can solve *tall* linear systems, giving the least squares solution *without having to use the normal equations*.
- QR factorization is numerically extremely well-behaved. Works even for singular (or zero-valued!) matrices.

Key idea

Geometric interpretation : rotate, project, then unwarp! $\mathbf{Q}, \mathbf{R} = \mathrm{qr}(\mathbf{A})$; $\mathbf{b}' = \mathbf{Q}^T\mathbf{b}$ gives a simpler *square* 2D problem; then $\mathbf{x} = \mathbf{R}^{-1}\mathbf{b}'$.

Regularization

Definition

REGULARIZATION
Methods that *improve the condition number* of a problem.

- There are many different ways to do so.
- No free lunch : better conditioning will come at the cost of decreased accuracy, bias, etc.

Definition

TIKHONOV REGULARIZATION
Assuming that a symptom of an unstable solution is that the solution $\mathbf{x}$'s magnitude becomes too large, penalize solutions with large L_2 norms. The regularization factor λ controls the degree to which we prefer smaller solutions.

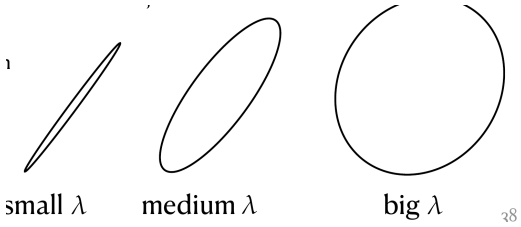
Key formulas

$$\begin{aligned}&\operatorname{argmin}\left[\mathbf{x}^T\mathbf{A}^T\mathbf{A}\mathbf{x} - 2\mathbf{b}^T\mathbf{A}\mathbf{x} + \|\mathbf{b}\|_2^2 + \lambda\|\mathbf{x}\|_2^2\right]\\ =\ &\operatorname{argmin}\left[\mathbf{x}^T\mathbf{A}^T\mathbf{A}\mathbf{x} - 2\mathbf{b}^T\mathbf{A}\mathbf{x} + \|\mathbf{b}\|_2^2 + \lambda\mathbf{x}^T\mathbf{I}^T\mathbf{I}\mathbf{x}\right]\end{aligned}$$

Trick : merge the identity into $\mathbf{A}^T\mathbf{A}$ and then just use the normal solver.

Key idea

Geometric interpretation : the addition of an identity matrix causes the matrix to scale input vectors more uniformly in different directions (small λ : thin ellipse; big λ : almost a circle).



38

Summary

Technical details

- **LU factorization** : simply add $\lambda\mathbf{I}$ to $\mathbf{A}^T\mathbf{A}$ (via normal equations)... and solve!
- **QR factorization** : factorize the extended matrix $\begin{bmatrix}\mathbf{A}\\ \sqrt{\lambda}\mathbf{I}\end{bmatrix}$ instead of $\mathbf{A}$, append n zeros to the right hand side $\mathbf{b}$.

Weekly Summary

Summary

- Normal equations : $\mathrm{cond}(\mathbf{A}^T\mathbf{A}) \approx \mathrm{cond}(\mathbf{A})^2$.
- Orthogonality fixes conditioning : Gram-Schmidt, Legendre polynomials; Chebyshev points against the Runge phenomenon.
- $\mathbf{A} = \mathbf{QR}$ (Householder reflections) : least squares via $\mathbf{Rx} = \mathbf{Q}^T\mathbf{b}$, without $\mathbf{A}^T\mathbf{A}$.
- Tikhonov regularization : $+\lambda\|\mathbf{x}\|_2^2$, better conditioning at the cost of bias.